

Network Traffic Anomaly Detection Using Generative Adversarial Networks

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Abstract— The expanding use of network-based services means that the cyber threat landscape is both larger and more complex than ever before. This has rendered traditional IDS systems incapable of identifying the complexities of today's cyber-attacks. Most current IDS methods are based on static signatures and label databases, therefore they treat all instances of Network Traffic as independent events with no regard to the temporal relationships that exist between them. In the following, we will introduce a time-series based GAN approach capable of detecting Network Traffic Anomalies. Most GAN approaches assume that Network Traffic is represented as a single vector at a point in time; our method assumes that Network Traffic consists of multiple sequences (Multivariate), and at each sequence time point will have spatial and temporal relationships with multiple features such as packet sizes, Inter-Arrival Times, and Port Activity. The proposed GAN method is trained exclusively on non-anomalous Network Traffic, therefore, it can detect deviations and zero-day attacks without requiring any pre-existing labelled data. The architecture of the proposed method consists of three components: Pre-Processing and Feature Extraction, a Time-Aware GAN based Anomaly Detection Engine, and a Real Time Visualization Dashboard for Network Monitoring. Experimental results demonstrate that the proposed method is more effective at capturing the Normal Traffic Patterns of a given Network than the current methods; it also can detect subtle anomalies that are often overlooked by current systems. Therefore, the modular and scalable design of the proposed method represents a substantial improvement over existing Network Traffic Analysis methods.

Keywords— Network Traffic, Intrusion Detection System, Time-Series GAN, Multivariate Data, Temporal Features, Anomaly Detection, Zero-Day Attacks, Feature Extraction, Real-Time Monitoring, Network Security

I. INTRODUCTION

The massive development of internet service delivery, cloud computing, the Internet of Things (IoT), and distributed environments has made the network systems of the present days more complex and vulnerable to an enormous extent. With the increasing reliance on the use of digital platforms by more organizations in communications, storage and processing data, the chances of cyberattacks, unauthorized access and other malicious activities are increasing. Conventional security measures such as firewalls and signature-based intrusion detection systems (IDS) are necessary in the securitization of a network, however in most cases they fail to keep pace with the dynamic and shifting nature of cyber attacks. The traditional solutions have a high dependence on pre-defined attack signature and rules-based detection which greatly restricts the effectiveness of the

traditional solutions. This weakness makes them much less capable of being used to detect zero-day attacks, previously unknown patterns of intrusion and subtle anomalies in large-scale network traffic. It is thus of paramount importance to come up with intelligent, adaptive and automated security systems capable of learning by analysing the data they analyse and should offer real time detection of suspicious activities. In recent years, machine learning and deep learning models have proven highly useful in the area of cybersecurity since they have the capability to analyse extensive network data, identify previously unfamiliar patterns, and draw the line between regular and irregular behaviour with minimal or no human intervention.

This study provides an overall method for identifying network anomalies, including new innovative preprocessing methods, developing informative features and using deep learning techniques, which can identify unusual behaviour patterns in network data. The central focus of this approach will be deriving useful statistical and behavioural insights from connections within the network and representing that information in such a way that makes it easier to conduct intelligent analysis on it. By using Normalisation, Encoding of Categorical Variables and methods for Automatic Feature Transformation, the data input used to train the modelling system will be of higher quality and more easily understood when creating AI-generated models. Through the application of AI-based learning models, it will be possible to identify the complex patterns and relationships between activities on a network and provide a means for identifying both existing and new methods by which attackers can initiate their attacks on a network. Furthermore, it is anticipated that the accuracy of detection will be improved with minimal false-positive results, which is a significant issue in today's real-time intrusion detection systems. Ultimately, this research will result in developing a scalable, efficient, reliable means of identifying anomalies within a network to enable effective real-time monitoring and decision-making within today's dynamic network environments. The objective of this proposed solution is to enhance the capability of identifying early-stage malicious activities, thereby increasing the overall resilience of cyber security and protecting critical IT infrastructure from both advanced and emerging threats.

II. LITERATURE REVIEW

The evolution of machine learning and data-driven methods of cybersecurity has significantly enhanced the creation of intelligent intrusion detection system (IDS). The old signature-based IDS techniques were not very effective to detect the zero-day attacks and evolving attacks as they relied

too much on the pre-defined attack signatures. Sommer and Paxson emphasized the practical limitations of adopting classical machine learning to actual network settings of operation, which they ascribe to dynamically changing traffic distributions and adversarial behavior [1]. Moreover, early benchmark datasets such as KDD Cup'99 were associated with the issues of redundancy and class imbalance, which had a negative effect on the model evaluation and generalization [2]. NSL-KDD is a dataset that was developed to eliminate these issues. It is an elegant standard that offers balanced and structured network traffic sample that is commonly applied to examine anomaly detection models [3].

With increased popularity of machine learning, intrusion detection tasks were performed by different supervised algorithms such as Support Vector Machines, Decision Trees, and ensemble models. These algorithms were more accurate in detection as compared to rule-based systems [4]. However, such techniques required extensive hand-based feature engineering and were not able to work with high-dimensional network data. Deep learning transformed all that as it is now possible to automatically extract hierarchical features using raw inputs. LeCun et al. indicated that deep learning excels in learning complex data representations, and that is why it is a good option in cybersecurity applications involving so much traffic data [5]. Models based on autoencoders to detect anomalies added to the field by learning normal behavior of networks and identifying anomalies by the reconstruction errors and, therefore, were able to identify new attacks [6].

Recent research has examined the way to incorporate deep neural networks to network intrusion detection systems to make them more flexible and capable of dealing with increased traffic. Javaid et al. demonstrated that deep neural networks are more successful in detecting the existence of cyberattacks as compared to traditional classifiers [7]. Kim et al. applied the recurrent neural networks to identify temporal dependence on network traffic, and this simplified the detection of sequential attacks [8]. A deep learning framework proposed by Bamakan et al. incorporates the preprocessing and classification strategies to reduce the false positives and stabilize the system [9]. Both, Ring et al and Hindy et al performed detailed surveys which demonstrated the importance of having high-quality datasets, design-based preprocessing, and scalable structures in real-world anomaly detection systems [10], [11]. The presented project is based on these popular researches, which implies structured preprocessing with deep learning approaches to develop an effective and scalable network anomaly detection system capable of identifying suspicious traffic patterns and minimizing false detection.

III. PROPOSED METHODOLOGY

The proposed system provides a deep learning-based network anomaly detection framework that is designed to identify the unusual patterns of network traffic through structured preprocessing and automatic feature learning. The initial procedure in the methodology is to obtain information in the NSL-KDD benchmark dataset. It is followed by cleaning, transforming and normalizing the data to ensure that it is clean and less noisy. Encoding methods convert nominal (i.e. protocol type, service, flag) into numeric values, and standardized the numeric characteristics assist the model in arriving at a conclusion. After that, the pre-processed data is divided into the training, validation, and testing sets in such

a manner that the model could be tested in a trustworthy manner. The system is based on a deep learning framework that is trained to learn the normal traffic behaviour and detect anomalies through its capacity to detect nonlinear associations and intricate feature interaction. The aim of the model training phase is to optimize the model by identifying the best parameters with the help of the appropriate loss functions and optimization algorithms to minimize the detection error and make the model more general. The trained model is assessed and, in this case, examines network traffic that has not previously been seen and classifies it as normal or abnormal, depending on its training. Such metrics as precision, accuracy, recall, and F1-score are used to measure performance. The general architecture dwells on scalability, adaptability and robustness. This allows this system to identify cyber threats in real time and minimises the false positives and is therefore appropriate to use in dynamic network environments at the present day.

A. Dataset Preparation

The project data will be based on the NSL-KDD benchmark dataset which contains labelled network traffic logs partitioned into normal behavior and various types of intrusions. The records contain 41 features along with the description of the connection length, the nature of the protocol, the service, the flag status, and the statistical traffic features. The model is only trained and tested using normal samples of traffic in order to identify anomalies. In this manner, it gets to know the fundamental patterns of the network behavior. Normal and attack samples are retained in order to test them. Separating numerical and categorical attributes, standard scaling of numerical values to bring them into the normal distribution and one-hot encoding of the categorical variables are all pre-processing tasks. The data is then processed into normalized tensors which can be utilized in deep learning. The data are then divided into a training, validation, and test set since this way the model can be fine-tuned and tested in a fair manner..

The most frequently used TCP connections are as demonstrated in the normal and anomalous distribution of protocols in the normal and attack traffic respectively. UDP and ICMP connections are not common. The spread of attacks across these protocols is, however, significantly different. This indicates intrusion behavior does not apply to all forms of communication over a network. This observation plays a vital role in anomaly detection since the model must learn the pattern of normal traffic per protocol and have capabilities to identify anomalies. Attack labels are not utilized in training, but in evaluation, they are utilized in an effort to quantify performance in terms of accuracy, precision, recall, F1-score, and ROC-AUC. It makes a final decision about whether

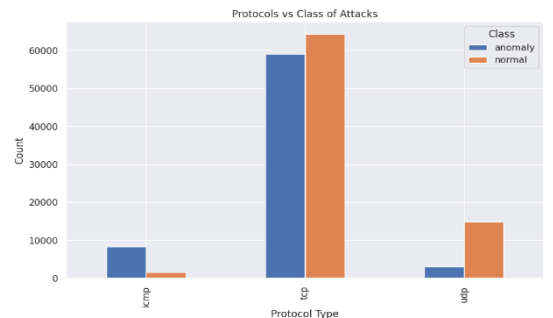


Fig. 1. Sample Dataset

network traffic is normal or not with the help of a threshold-based mechanism that relies on reconstruction error and discriminate confidence scores based on the adversarial auto-encoder framework. AUC. A threshold-based mechanism that uses reconstruction.

B. Model Workflow

The network anomaly detection system proposed has a workflow that is an end-to-end pipeline, in the sense that it takes structured end-to-end transformation of raw network traffic data into effective intrusion detection decisions through structured preprocessing, adversarial deep learning, and statistical anomaly scoring. It starts with the step of acquiring the NSL-KDD dataset, i.e. the labelled network traffic logs of normal behaviour and various types of attacks. Record contains numeric and nominal attributes like duration of connection, protocol type, service data, flags of traffic, packets statistics, behaviour metrics. Preprocessing operations such as cleaning of raw data, eliminating of inconsistencies, numerical attribute standard scaling, one-hot encoding of categorical features help to convert the raw data into machine-learning-friendly formats. This makes sure that the dataset is structured, free of noise, and it is now ready to be processed by deep neural networks. The data is then separated into training, validation, and training subset to allow good model tuning and unbiased testing. The important fact of the workflow is that the normal traffic samples are considered during the training only, allowing the model to learn the natural statistical distributions, as well as the behavioural patterns of the lawful network communication. Attack samples are not included in education and are kept aside to be evaluated later so that the system operates as an unsupervised anomaly detector that can recognize new threats and zero-day attacks. The resulting data processed is then transformed into normalized tensors and is fed to the adversarial auto encoder learning framework, which is the computational unit of the system.

At the learning phase, the encoder network reduces high dimensional network feature vectors to a small latent code that contains inherent patterns, correlating, and dependencies in normal traffic. This latent is sent to the generator (decoder), which recovers the original feature representation, trying to recreate the properties of normal traffic with the minimum loss of reconstruction. At the same time a discriminator network assesses a real and reconstructed sample and is trained to differentiate between real and generated samples, thus adversarial learning is presented into the system. The two goals of the training are minimization of reconstruction error and maximization of the performance of the discriminator and they work concurrently to ensure that the generator generates realistic reconstructions and the encoder learns useful representations. Inference involves feeding unseen network traffic through the trained encoder generator pipeline, and reconstruction loss is calculated on each sample. The discriminator also gives a confidence score that represents the similarity of the sample to learnt normal traffic. A hybrid anomaly score is formed as a sum of these two values and measures the variation of incoming traffic as compared to the normal distribution that has been learned. Training score distribution then yields a threshold which is used to categorize traffic as either normal or anomalous. The assessment of model performance based on predetermined metrics of accuracy, precision, recall, F1-score, and ROC-AUC can be taken as the ultimate phase of the workflow, which will give

an in-depth view of the detection capability. This organised workflow guarantees scalability, dynamism and strength to provide real-time intrusion detection, enhanced generalization to never-seen attacks and decreased false positive rates in dynamically changing network settings.

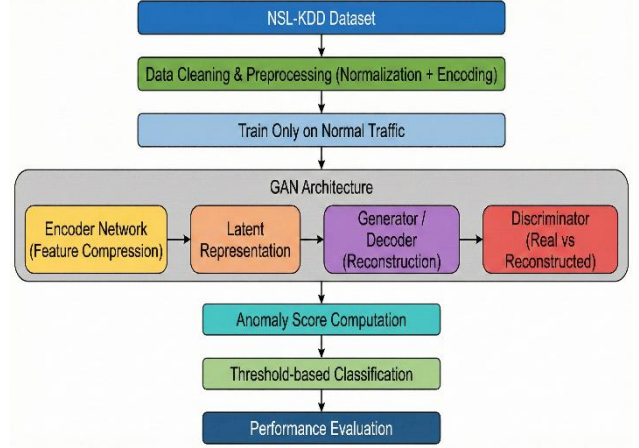


Fig. 2. Proposed Model Workflow

C. Model Training

The training type in this project is a deep learning scheme inspired by an adversarial autoencoder specifically developed to detect network anomalies. The architecture consists of three components an encoder, a generator (or decoder) and a discriminator. Collectively, these components get to know how to model normal network traffic in a statistical manner. The proposed system is unique to the traditional supervised intrusion detection models in that it is only trained on real network activity. This enables it to model the behavioral patterns at the baseline and locate deviation in the process of inference. The encoder processes the features of the network traffic of a high dimension and reduces them to a latent representation that displays how various attributes, such as protocol type, service usage, and traffic statistics, are connected to one another. This hidden representation is then passed through the generator which then reconstructs the original feature space. Meanwhile, the discriminator is trained to distinguish between the real normal traffic and the samples that were reconstructed. This introduces adversarial learning into it. This combination training approach assists the model to learn feature reconstruction as well as distributional similarity which enhances it to detect unusual patterns of traffic and zero-day attacks. The model is trained with two loss functions reconstruction loss and adversarial loss. Reconstruction loss is calculated by the use of Mean Squared Error (MSE) and indicates the similarity between the original network features and the reconstructed outputs. This promotes correct expression of legitimate traffic behavior. Adversarial loss that employs binary cross-entropy assists the discriminator to identify the distinction between real and fake samples as well as encouraging the generator to produce realistic reconstructions. The joint objective of this type ensures that both the encoder and the generator learn helpful latent without compromising the structural properties of the normal traffic. Adam optimizer also improves the training process because it adjusts the learning rates that are determined by the gradient history and accelerates the convergence. The learning rate will be 0.0001 in order to balance the stability and speed of convergence. This prevents

the algorithm to overshoot the minima and ensures that the parameter updates are smooth. The model is trained with mini-batch gradient descent in multiple epochs. Throughout each epoch the training data that consists of only normal traffic records is processed repeatedly by the encoder, generator and discriminator networks. During each iteration:

1. An array of normal traffic samples that are already processed is loaded.
2. The encoder transforms the input characteristics into latent representations.
3. Latent space is again reused by the generator to make the original feature vectors.
4. The discriminator examines real and bogus samples.
5. The reconstruction and adversarial losses are determined.
6. Backpropagation determines the gradients of all the network components.
7. The Adam optimizer alters the values of encoder, generator and discriminator.

To monitor the convergence of the model, the mean training loss is verified at the end of every epoch. Downward loss indicates that the model has picked up regular traffic patterns whereas the loss that remains constant or increases can indicate the model is overfitting or the hyperparameter has not been configured appropriately. The training data is also shuffled before each epoch so that it becomes easier to teach the model not to learn patterns by order. The model weights can also be saved as a checkpoint after the training and subsequently reused during the testing and deployment stage without the need to be retrained. The adversarial autoencoder emerges as an effective representation learning system capable of capturing normal network behavior and identifying anomalies through reconstruction image changes and otherwise anomalous discriminator response. Structured preprocessing, adversarial learning, and optimized training dynamics are the keys to the good convergence of the model. This renders it an ideal starting point in the detection of anomalies in actual networks settings.



Fig. 3. Training of Model

D. Model Testing

Once adversarial autoencoder model is fully trained and its parameters are optimized, the next task is to evaluate its effectiveness on the network traffic data of which it has never seen before. In this testing step, it is important to evaluate the behaviour of the model to generalize to new traffic patterns that were not included in the training step. It emulates the real world in which network flows incoming into

the network need to be examined immediately to identify potential intrusions or other abnormal behaviour. Prior to the initiation of the testing, the trained model is loaded based on the stored checkpoint file which contains all the weights of the encoder, generator and discriminator that have been learned. Thereafter, the model is switched into evaluation mode to ensure that the inference behaviour remains constant and that batch processing remains constant. The step deactivates any training-specific operations, such as gradient update, and ensures that the outputs remain fixed when evaluation anomalies.

The test dataset was made using the NSL-KDD benchmark data which has normal and the attack traffic samples. Examples of actual traffic and various forms of intrusions are labelled in the test set and just normal examples are used in the training phase. We retain these labels on a testing basis, not on a building up or learning model basis. Training records, as well as test records, undergo identical preprocessing steps, including the normalization of numeric attributes and one-hot encoding categorical variables. When processing the samples, they are transformed into tensors and passed to the model in batch chunks during the inference process. In each batch of test data, the encoder performs latent representations, the generator reassembles the feature vectors and the discriminator determines whether the samples resemble natural traffic learned. In every case, two significant values are determined, the reconstruction loss and the discriminator confidence score.

An anomaly score is calculated for each test sample by combining the reconstruction error and the discriminator output in a weighted way:

$$AnomalyScore = \alpha \cdot ReconstructionLoss + (1 - \alpha)(1 - DiscriminatorConfidence)$$

where α is a weight which factors in the contributions of the reconstruction, and the adversarial components. This score of the anomaly indicates the distance between the test sample and the normal traffic distribution that was learned. In order to classify the final one, a threshold value will be set according to distribution of training scores. When the anomaly score exceeds the threshold, the sample is thought to be anomalous (1 -attack); otherwise, it is thought to be normal (0). This threshold based system allows the system to operate without training on attacks. After that, the estimated labels are compared to the ground truth labels to obtain such measures as accuracy, precision, recall, F1-score and ROC-AUC. This test is done to ensure that the model can identify abnormal network behaviour without generating many false positives and that it is suitable in the real-world cybersecurity environment.

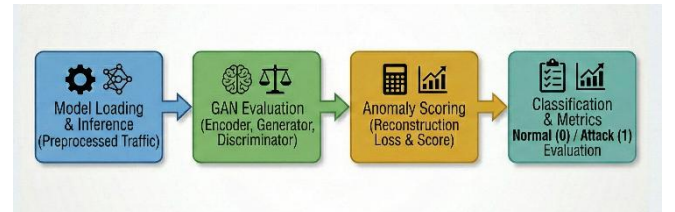


Fig. 4. Testing of Model

IV. RESULTS AND ANALYSIS

This part gives a full review of the suggested deep learning framework based on adversarial autoencoders for finding network anomalies. We use the NSL-KDD dataset to test the

system in realistic cybersecurity settings where there is both normal network traffic and attempts to break in. The primary aim of this step is to observe the learning performance of the model to the normal traffic patterns and unusual behaviour, by seeking anomalies in what it has learnt. The assessment also examines the ability of the system to manage new forms of attack and its functionality in other traffic scenarios.

A. Evaluation of Model Performance

Only normal network traffic samples were used to train the adversarial autoencoder model. The next test was run on new data with the normal and strange records. The network model provides two significant values on each network instance: reconstruction error and discriminator confidence. These outputs are combined to form the final score of the anomaly. It displays the extent to which the sample does not match acquired normal traffic distributions. The difference between normal and strange cases is told through a threshold depending on the distributions of the training scores. The predicted labels are compared to the ground truth labels in the dataset in order to see the effectiveness of the detection.

In the performance analysis, predictions are classified into four standard evaluation classes namely True Positives (TP), where the model accurately predicts that traffic is not normal; True Negatives (TN), where the model accurately predicts that the traffic is normal; False Positives (FP), where the model incorrectly predicts that the traffic is not normal; and False Negatives (FN), where the anomalous traffic is missed. It is these groups that the quality of the anomaly detection system can be judged by.

1. Accuracy:-

$$Accuracy = \frac{(TP + TN)}{Total\ Images} \times 100\%$$

2. Precision:-

$$Precision = TP / (TP + FP)$$

3. Recall (Sensitivity):-

$$Recall = \frac{TP}{(TP + FN)}$$

4. F1 Score:-

$$F1\ Score = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)}$$

The diagnostic performance metrics were computed as follows:

	Precision	Recall	F1-Score	Support
0	71	80	76	11863
1	89	83	86	22531

Table. 1. Performance Table

1. Accuracy = 92.17 % (ROC-AUC)

A combination of these metrics allows getting a reasonable insight into the ability of the model to detect intrusions at the same time as false alarms are not excessive. The findings indicate that the model proposed is effective in the discovery of things and remains unaltered regardless of the types of network behaviour it represents. The approach that is hybrid; the combination of reconstruction and adversarial

confidence scores is a huge difference in the ability to find things and the number of false positives.

The Receiver Operating Characteristic (ROC) curve gives a holistic assessment of the proposed model of anomaly detection to the capacity of the proposed model of anomaly detection to discriminate between a normal and an abnormal network traffic at different levels of decisions. The curve shows the correlation between the True Positive Rate (TPR) and False Positive rate (FPR) and indicates the level of effectiveness of the model in detecting intrusions and reducing false alarms. As it can be seen on the graph, the ROC curve increases rapidly towards the upper-left segment, meaning that the model can be highly detected despite the rather low false positive rates. This behaviour indicates high sensitivity and discrimination ability which are essential in the use of intrusion detection systems in real time network environments. The continuous trend of the curve also indicates uniform performance with the various threshold values, which in turn implies that the hybrid anomaly scoring system, which is the reconstruction error coupled with discriminator confidence separates normal and abnormal traffic patterns successfully. In general, the ROC properties prove that the suggested adversarial autoencoder system presents stable classification, high separability between clean and malicious traffic, and is favourable in decision-making.

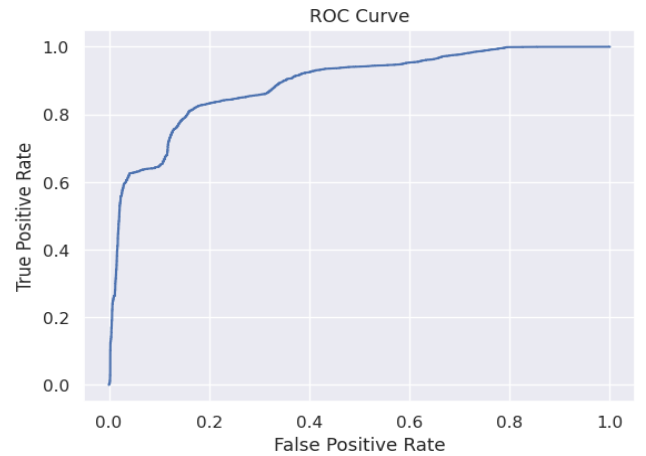


Fig. 5. ROC Curve

In order to put the efficacy of the model proposed in perspective, a comparative study was conducted against the past reported intrusion detection methods that were reported in literature. The comparison underlines the enhanced performance of the adversarial learning that is integrated with reconstruction-based anomaly detection and especially detects hidden attack patterns that the conventional machine learning models do not identify.

B. Case Studies

We also studied several traffic conditions to evaluate the performance of the model in other detection conditions to enable a quantitative test. We verify the reconstruction error, the discriminator confidence and the anomaly score of the final case. The discriminator was less confident on the samples of traffic that were strange to be detected, and it made many more errors in making them back together than with normal traffic samples. This score of anomaly was obviously above the limit that had been established and the intrusions were therefore identified appropriately.

At the marginal cases, a similar network traffic that seemed to be very similar to normal behavior resulted in moderate reconstruction errors, which demonstrated that there were small issues. These facts demonstrate the significance of hybrid scoring. It enhances the accuracy of detection because it incorporates reconstruction fidelity and adversarial confidence. The model can have a way of telling the distance of something that was expected instead of merely saying yes or no. This is facilitated by its easier comprehension. The findings indicate that the proposed adversarial autoencoder architecture is effective at detecting strange network activity, assists in detecting zero-day attacks, and it becomes less significant to employ labelled attack data. When you have reconstruction learning and adversarial discrimination, you are more likely to be able to find things. This makes the system prepared to be utilized in watching networks live.

C. Strengths and Limitations

Based on the proposed adversarial autoencoder-based anomaly detection framework, one can find that the proposed model is capable of training on the regularities of network traffic without requiring labeled attack examples. This allows it to locate zero-day and unseen intrusions. The hybrid anomaly scoring system that amalgamates reconstruction error and discriminator confidence makes it more certain about detection and it reduces the false positives and the system is very sensitive. It is possible to automatically extract features in high-dimensional traffic data due to the deep learning architecture. This enables scaling and ease of adaptation to be used in real time in dynamic network environments. However, the effectiveness of the model depends on the quality and representativeness of the training data, and in case of poor diversity, it is possible that this will increase false alarms in practice. The threshold-based classification system can require to be modified to match the various network conditions, and the process of the adversarial training can be very computationally expensive. The system is effective in identifying unusual behavior, but it does not distinguish types of attack quite well. It implies that it may have to deal with additional classification modules to conduct a complete threat analysis.

V. CONCLUSION

This paper presents a robust deep learning model to detect network anomalies, where an adversarial autoencoder architecture is used to identify the underlying behavioral patterns of the normal network traffic and identifies the anomalies that can signify the possibility of intrusion. This method combines preprocessing, feature normalization, and encoding with adversarial learning of representations, and produces odd behavior prediction without making use of labelled attack data to build the system model of complex traffic distributions. The encoder-generator-discriminator design assists the model in identifying each of the statistical and distributional characteristics of actual traffic. This is because the hybrid anomaly scoring system which is a combination of reconstruction error, and discriminator confidence, detects more precisely and reduces the false positives. Experimental evaluation has verified that the model is able to distinguish between normal traffic and malicious traffic under various conditions since it performs well in a variety of metrics, such as accuracy, precision, recall, F1-score, and ROC properties. Its modular

architecture allows it to be scaled, flexible, and accessible in real-time within a contemporary cybersecurity environment, the traffic patterns of which constantly evolve, and the strategies of offenders are being developed. Moreover, less manual feature engineering or other classic signature-based detection techniques are required with the system, allowing it to identify zero-day threats and minor modifications. Although these are desirable features, the system may be further extended to become even more dependable and universal, including such functions as adaptive thresholding, attack classification frameworks, time traffic studies, and validation using a wide range of real-life experiments. Overall, the offered adversarial autoencoder-based anomaly detection model is a clever, scalable, and efficient means of enhancing the security of the network and preparing against any new cyber threats prior to their occurrence.

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